

Gerbv: The Gerber Viewer

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The period between design completion and printed circuit board (PCB) manufacturing can be tricky. First, you need to completely validate the layout you have created. Once you have finalised it but the fabrication unit finds faults, the layout has to be thoroughly analysed, errors have to be spotted and corrected, and any other suggestions to be implemented, in order to get the board you dreamt of.

Gerbv is a tool that lets you view, print and edit Gerber files—the ones that act as an input for producing a PCB. You can analyse and validate your file's syntax, code count, aperture usage and other such factors. This could be your solution to problems during the computer-aided manufacturing stage of your board.

Input plenty, output plenty as well

The file sent in to Gerbv could be in formats like RS-274X, Excellon drill and pick-place. Pick-place formats contain x and y axes information, rotation details and placement side, and are of comma-separated ASCII (.csv) type. Your job is made easier as the tool automatically reads and detects the file format when multiple formats are fed as inputs.

These files are typically generated by computer-aided manufacturing export function of your layout tool. Once generated, this tool is one of your best choices to manipulate its details. When you are done with all the viewing and editing, Gerbv lets you export the files in formats suited to your application. Supported formats include the very popular

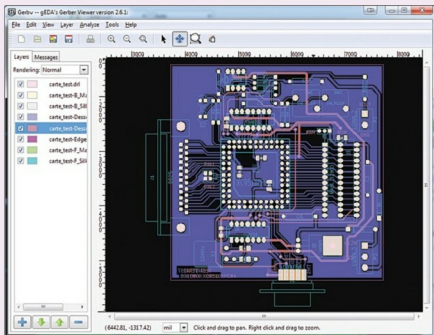


Fig. 1: A PCB for your IC

What you need to know about Gerber

A Gerber file is typically the output from a PCB computer-aided designing tool, the one that is sent for manufacturing your PCB. It generally follows one of the two standards, RS-274D or the more recent RS-274X; main difference being inclusion of aperture format.

Each layer of your design, like top silk and metal layers, is stored in separate files, all of which have to be included in Gerbv for a wholesome picture. There are many other tools that deal in Gerber files like GerbvQt, Pcb2gcode, Gerber2tiff and GBTiler, to name a few.

png or pdf formats, and even ps or svg file types.

A layered approach

As you progress into better and more complicated layouts, the first noticeable change is the increase in number of layers; the manufacturer permitting the same, of course. The moment this comes into picture, so does design complexity.

Needless to say, verification is a challenge. The best way is to strip down the design layer by layer and

study the numerous traces zig-zagging everywhere.

This is something you can do with ease using Gerbv. You can view layers individually, combine the selected few layers and even change layer polarity for negative or positive display. The layers can also be stacked up in any order as you please, by simply dragging and dropping these—definitely a boon when trying to analyse traces travelling over multiple layers.

With Cairo, you can make the